The Baseline Tension: Solar Coherence

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We present interferometric observations of the Sun at 10.45 GHz obtained with the two-dish correlating interferometer at New Campbell Hall, Berkeley. From a horizon-to-horizon drift scan comprising 8226 visibility integrations across 285 spectral channels, we recover the baseline geometry, instrumental delay, and source structure. Five independent baseline-determination methods: STFT fringe-frequency fitting, phase-slope regression, delay-spectrum analysis, nonlinear least-squares fringe fitting, and brute-force grid search yields an adopted baseline of $b_{\rm ew} = 15.11 \pm 0.01$ m and $b_{\rm ns} = 1.49 \pm 0.05$ m, consistent with an independent LiDAR measurement at the 0.3σ level. The east–west component is tightly constrained across all methods, while the north–south component shows greater scatter due to its degeneracy with the geometry and cable delay. The angular diameter of the Sun, measured from the Bessel-function modulation of the visibility envelope, is $D_\odot = 34.66 \pm 0.03'$ (extrema) and $34.46 \pm 0.01'$ (jinc fit), both $\sim$5% above published radio values at comparable frequencies; the excess is attributed to systematic baseline uncertainties rather than a physical chromospheric effect. Non-zero visibility at the Bessel nulls is consistent with an unresolved active region on the disk, confirmed by SDO/HMI imagery from the observation date.

1. Introduction

Radio interferometry is the foundation of modern radio astronomy and the technique by which the event horizon of a supermassive black hole was first imaged. By correlating the signals from two or more spatially separated antennas, an interferometer achieves angular resolution not of the diameter of any single dish, but by its baseline separation.

In this report, we use the two-dish X-band interferometer at the New Campbell Hall (NCH) to explore three scientific goals: **(1)** calibration of the baseline vector $(b_{\rm ew}, b_{\rm ns})$ via 5 independent methods; **(2)** measurement of the solar diameter, which we expect to contain emissions beyond the optical photosphere into the chromosphere; and **(3)** indirect detection of unresolved sunspot emissions as a non-zero visibility amplitude floor at Bessel-function nulls.

2. Background Theory

A single radio dish of diameter D has an angular resolution diffraction-limited to $\theta \sim \lambda/D$. At our observing wavelength of $\lambda = 2.87$ cm, a 1-metre dish resolves only $\sim$1.6° which would be too coarse to resolve the Sun's $\sim$0.5° disk, much less finer structures. A 15-metre dish would be needed to match the Sun's angular size, and arcsecond resolution would require a dish hundreds of metres across.

Interferometry circumvents this; instead of building one enormous aperture, we place two dishes a distance b (baseline) apart and correlate their signals. The correlated output is sensitive to angular structure on the scale $\theta \sim \lambda/b$, equivalent to a full dish of aperture diameter b. For a 15-metre baseline at 2.87 cm, this corresponds to $\theta \sim 6.6'$, in the order of the solar diameter and sufficient to resolve any disk structure.

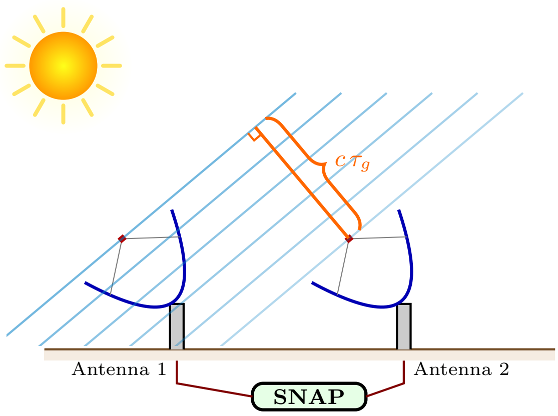

Fig. 1.— Illustration of a two-dish correlating interferometer. Two dishes separated by baseline b observe an extended source. Plane wavefronts arrive at the two antennas with a relative geometric delay. The signals are brought to the SNAP board, which digitises, Fourier-transforms, and multiplies the voltages to produce the complex visibility spectrum $V_{12}(\nu, t)$. As the Earth rotates, τ_g changes, and the correlated output traces out the interferometric fringe.

Two dishes separated by baseline b observe the Sun at an angle θ from the zenith. Plane wavefronts from the source arrive at the two dishes with a relative path delay τ_g, which depends on the projected baseline and changes as the Earth rotates. The signals are brought together in the SNAP board, which multiplies the Fourier-transformed voltages to produce the complex visibility spectrum. Figure 1 illustrates the geometry.

Rather than measuring brightness, an interferometer measures the complex visibility $V(u)$, which is the

Fourier component of the sky brightness distribution at the spatial frequency $u = b_{\rm proj}/\lambda$, where $b_{\rm proj}$ is the projected baseline perpendicular to the line of sight. Here, $b_{\rm proj} = \sqrt{|\mathbf{b}|^2 - (\mathbf{b}\cdot\hat{\mathbf{s}})^2}$, where $\hat{\mathbf{s}}$ is the unit vector toward the source. The component along the line of sight, $\mathbf{b}\cdot\hat{\mathbf{s}} = c\,\tau_g$, is the geometric delay.

From a fixed position, the projected baseline changes continuously as the Earth rotates, and an interferometer observing an object samples a different spatial frequency. Over a full horizon to horizon observation, u varies and a single baseline traces out an arc in the Fourier plane. By combining visibility measurements at many spatial frequencies, one can reconstruct the image of the source.

For a point source, the visibility amplitude is constant and the phase oscillates with the changing geometric delay producing fringes; for an extended source like the Sun, the visibility amplitude is modulated by the Fourier transform of its shape, producing characteristic nulls where the fringe spacing matches the source size (Section 2.4). The positions of these nulls corresponds to the Sun's diameter.

2.1. Path Delay

We decompose the baseline into an east–west component $b_{\rm ew}$ and a north–south component $b_{\rm ns}$. For a source at hour angle h and declination δ, observed from latitude L, the east–west baseline contributes a geometric delay

$$\tau_{g,\rm ew}(h) = \frac{b_{\rm ew}}{c}\cos\delta\,\sin h, \tag{1}$$

and the north–south baseline contributes

$$\tau_{g,\rm ns}(h) = \frac{b_{\rm ns}}{c}\sin L\,\cos\delta\,\cos h - \frac{b_{\rm ns}}{c}\cos L\,\sin\delta. \tag{2}$$

Since the second term is independent of h, we absorb this constant NS term into the instrumental cable delay τ_c, defining an effective geometric delay

$$\tau_g'(h) = \frac{b_{\rm ew}}{c}\cos\delta\,\sin h + \frac{b_{\rm ns}}{c}\sin L\,\cos\delta\,\cos h \tag{3}$$

and a modified cable delay

$$\tau_c' = \tau_c - \frac{b_{\rm ns}}{c}\cos L\,\sin\delta. \tag{4}$$

The total relative delay between the two signal paths is $\tau_{\rm tot} = \tau_g'(h) + \tau_c'$.

2.2. The Point-Source Fringe

For a monochromatic point source of frequency ν, the voltages at the two antennas are

$$E_1(t) = \cos(2\pi\nu t), \qquad E_2(t) = \cos\!\big(2\pi\nu\,[t + \tau_{\rm tot}]\big). \tag{5}$$

The aforementioned correlator output is proportional to $\langle E_1(t)\,E_2(t)\rangle$, the time-averaged product of the two signals. Together, the fringe output is

$$\begin{aligned} F(h) &= \cos\!\big(2\pi\nu\,[\tau_g'(h) + \tau_c']\big) \\ &= A\cos\!\big(2\pi\nu\,\tau_g'\big) + B\sin\!\big(2\pi\nu\,\tau_g'\big), \end{aligned} \tag{6}$$

where $A = \cos(2\pi\nu\,\tau_c')$ and $B = -\sin(2\pi\nu\,\tau_c')$ encode the instrumental phase. The unknown baseline parameters appear only inside τ_g', while A and B are linear and can be solved analytically, making this a separable nonlinear least-squares problem.

Physically, the interferometer projects a sinusoidal pattern onto the sky with fringe spacing $\lambda/b_{\rm proj}$, producing the oscillating visibility. Equivalently, the two antennas act as a giant double-sideband mixer, using the geometric delay rate to down-convert spatial structure as a temporal fringe. For extended sources, the output depends on how the source overlay the fringes: when the fringe spacing matches the source diameter, positive and negative lobes cancel and the visibility vanishes (Figure 2).

Fig. 2.— Fringe pattern projected on the sky for an extended source. *Left:* Short projected baseline produces wide fringes; the disk fits within one lobe and the visibility is high. *Right:* Long projected baseline produces narrow fringes; the disk spans multiple lobes whose + and − contributions partially cancel, reducing the visibility. At a Bessel null the cancellation is exact.

2.3. Fringe Frequency

The fringe phase $\phi(h) = 2\pi\nu\,\tau_g'(h)$ varies with hour angle. Differentiating the fringe phase with respect to time gives the instantaneous fringe frequency:

$$f_f(h) = \omega_\oplus\left[\frac{b_{\rm ew}}{\lambda}\cos\delta\,\cos h - \frac{b_{\rm ns}}{\lambda}\sin L\,\cos\delta\,\sin h\right], \tag{7}$$

where $\omega_\oplus = 7.2921\times10^{-5}$ rad s^{-1} is the sidereal rotation rate. For our baseline at $\delta \approx 0^\circ$, the fringe period at the meridian is $\sim$26 s. Since $f_f(h)$ varies slowly, a short-time Fourier transform (STFT) recovers it for every h bin, and fitting the measured $f_f(h)$ curve to equation (7) yields a baseline estimate (Method 1, §4).

For an extended source with brightness distribution $I(\Delta h)$, the interferometer response is the convolution of the point-source fringe with the source profile. For a symmetric source this factors as:

$$R(h_s) = F(h_s) \times \int I(\Delta h)\,\cos(2\pi f_f\,\Delta h)\,{\rm d}\Delta h. \tag{8}$$

where the integral adds a slow-varying modulation to the fringe pattern.

2.4. Uniform Disk Visibility

Modelling the Sun as a uniform disk of radius R, the modulating function evaluates to the jinc:

$$\frac{V(|u|)}{V(0)} = \frac{2\,J_1(2\pi\,|u|\,R)}{2\pi\,|u|\,R}, \tag{9}$$

with zeros at $|u|\,R = j_{1,k}/(2\pi)$, where $j_{1,k}$ are the zeros of J_1. Differentiating with respect to $|u|$, the extrema of $|V|$ satisfy

$$\frac{\mathrm{d}}{\mathrm{d}|u|}\left[\frac{V}{V(0)}\right] = -\frac{2\,J_2(2\pi|u|R)}{|u|} = 0, \tag{10}$$

so the minima (nulls) occur at the J_1 zeros $j_{1,k}$ and the maxima (sidelobe peaks) at the J_2 zeros $j_{2,k}$. Both provide independent radius estimates:

$$R = \frac{j_{n,k}}{2\pi\,|u_k|}, \tag{11}$$

where $j_{n,k}$ is the appropriate J_1 or J_2 zero for the k-th identified extremum.

3. Instrument and Observations

3.1. The X-Band Interferometer

The NCH interferometer consists of two dishes on the roof of Campbell Hall ($L = 37.8732°$ N, 122.2571° W), with a nearly east–west baseline. Each dish feeds a double-conversion heterodyne receiver: LO1 at 8.750 GHz converts the sky signal from the RF band (centred at 10.450 GHz) to an IF near 1.7 GHz via a DSB mixer, and LO2 at 1.545 GHz down-converts the IF to baseband via an SSB mixer. A K&L Microwave bandpass filter (5B120-155/110-B/B) at the output of each baseband chain limits the usable bandwidth to ∼70 MHz (sky: 10.415–10.485 GHz), yielding 285 usable channels (ch 513–798) out of the full 250 MHz Nyquist band.

A cross-correlator implemented on a SNAP signal-processing board (CASPER 2024) digitises both baseband signals at 500 Msps, computes a 2048-point real FFT (yielding 1024 spectral channels with spacing $\Delta\nu = F_s/N_{\mathrm{FFT}} = 244$ kHz), and multiplies the resulting complex voltage spectra. The cross-correlation is integrated over 305,200 spectra, corresponding to an integration time of 1.25 s per visibility.

Figure 3 shows a summary block diagram of the signal path for one antenna; the full detailed diagram is available online[1].

An independent baseline survey was performed using the LiDAR sensor of an iPhone 13 Pro, yielding $b_{\mathrm{ew}}^{\mathrm{lidar}} = 15.2 \pm 0.3$ m and $b_{\mathrm{ns}}^{\mathrm{lidar}} = 1.4 \pm 0.3$ m. The conservative 30 cm uncertainty encompasses the difficulty of identifying the true cardinal pointing of each baseline vector.

3.2. Observations

Horizon-to-horizon drift-scan observations of the Sun were obtained on 2026 March 20 (UTC). The data span hour angles from −79.9° to +22.0°, totalling 8226 visibility integrations divided across three observation CHIPS (Contiguous Horizon-to-horizon Integration Period Segments). The Sun's declination ranged from $\delta = -0.144°$ to $-0.032°$ over the observation. Table 1 summarises the observation log.

Table 1: Observation log.

CHIPS	Captures	HA range [deg]	Duration [hr]
0	5514	−79.9 to −8.0	4.8
1	403	−7.7 to −5.3	0.2
2	2309	−3.3 to +22.0	1.7

Channels at multiples of 256 (indices 0, 256, 512, 768) are contaminated by DC and FPGA pipeline harmonics and are excluded from all analysis. Residual ADC offsets in the SNAP board introduce a slowly

[1] https://github.com/AaronParsons/ugradio/blob/main/lab_interf/interf_signal_path.xcf

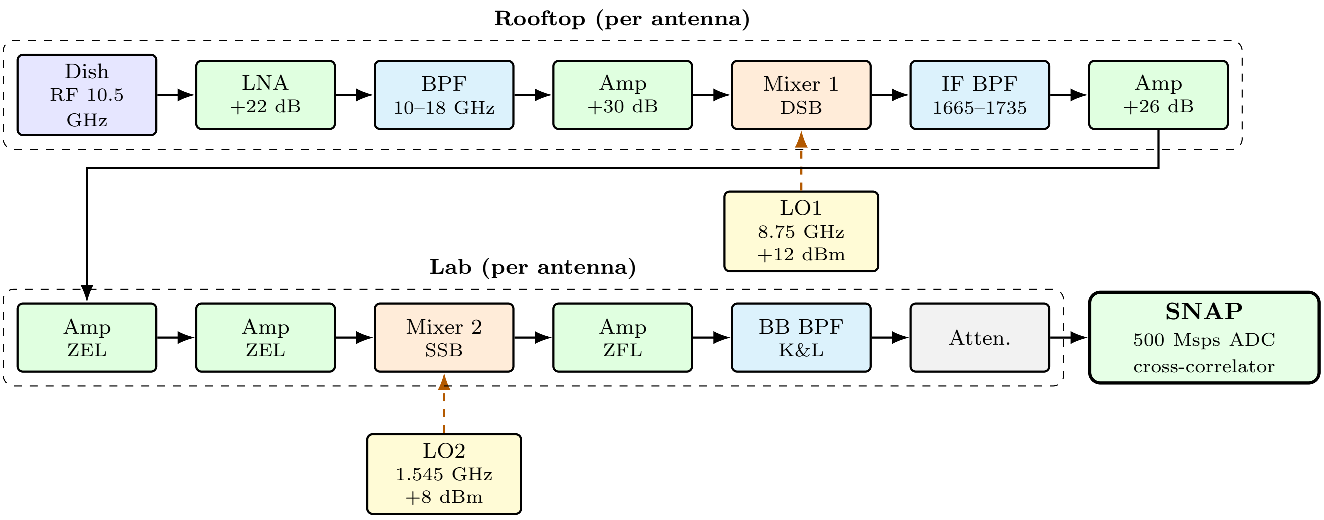

Fig. 3.— Summary signal path for one antenna of the NCH interferometer. The X-band RF signal (∼10.5 GHz) is amplified, filtered, and down-converted to IF (∼1.7 GHz) by DSB Mixer 1 with LO1 at 8.75 GHz. In the lab, SSB Mixer 2 with LO2 at 1.545 GHz converts to baseband (0–250 MHz) for digitisation by the SNAP board. Both antennas share the same LO1 and LO2 sources.

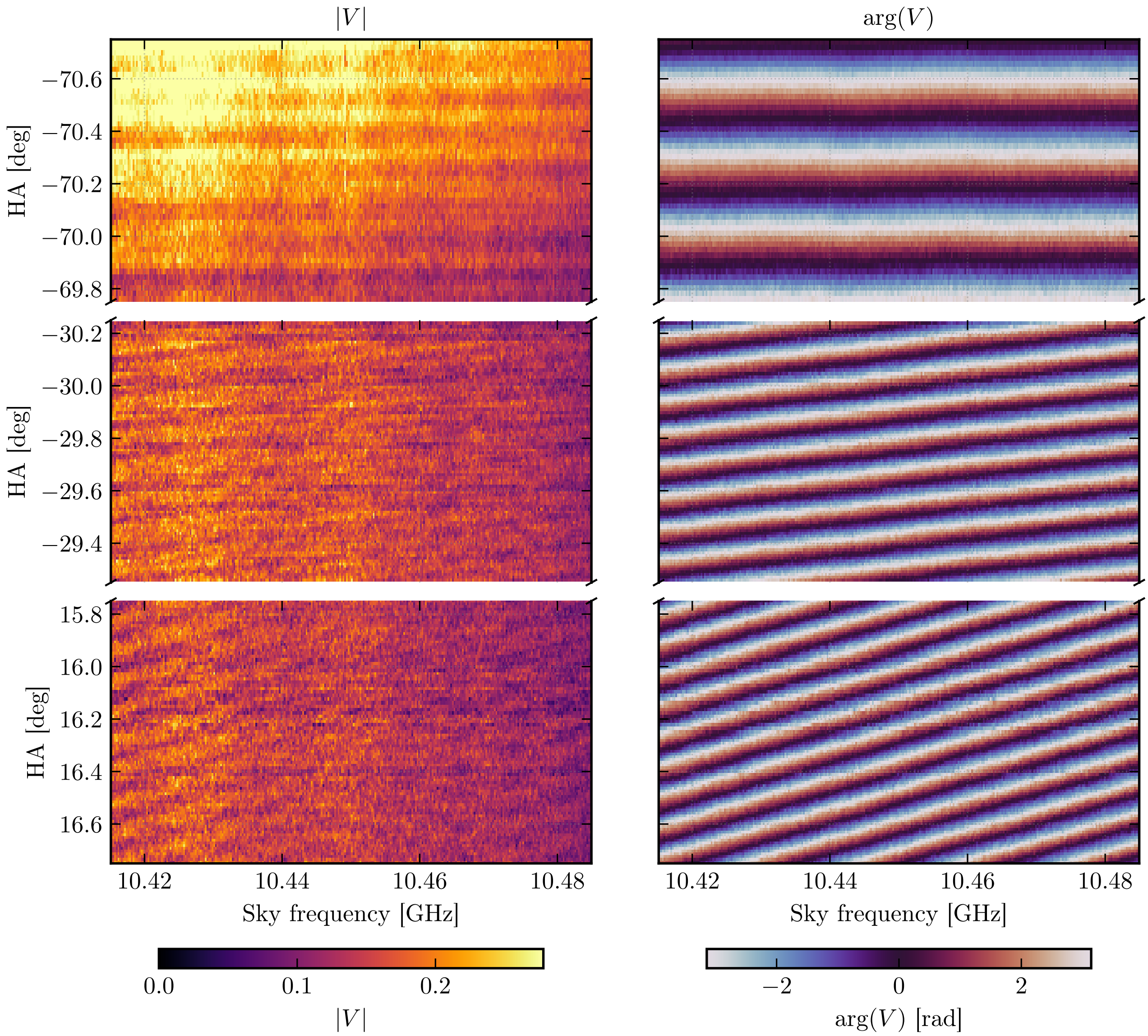

Fig. 4.— Waterfall plot of the DC-corrected visibility across the analysis band at three hour-angle windows. *Left columns:* Visibility amplitude $|V(\nu, h)|$. *Right columns:* Visibility phase $\arg V(\nu, h)$. The fringe pattern is tightly packed far from transit (high f_f) and spreads out near transit (low f_f). The phase slope across frequency at any given time recovers the geometric delay $\tau_g(h)$.

varying DC pedestal on $\mathrm{Re}(V)$ that must be removed before fringe analysis. Figure 4 shows the resulting visibility waterfall of the Sun across the analysis band at three hour-angle windows, displaying both the fringe amplitude and the phase slope across frequency that encodes the geometric delay.

3.3. DC Correction

We remove the DC pedestal by subtracting an adaptive rolling median whose window width scales with the local fringe period:

$$W(h) = N_P \times P_f(h) = \frac{N_P\,\lambda}{\omega_\oplus\, b_{\mathrm{ew}}\, \cos\delta\,|\cos h|}, \qquad (12)$$

where $N_P = 3$ fringe periods. This ensures the window always spans the same number of complete fringe cycles, regardless of hour angle, so the median is not biased by the fringe oscillation.

Figure 5 compares the raw and DC-corrected $\mathrm{Re}(V)$ at the representative channel. Figure 4 shows the DC-corrected visibility as a waterfall across the analysis band. The visibility phase at any given time varies linearly across frequency with slope $\mathrm{d}\phi/\mathrm{d}\nu = 2\pi\,\tau_g(h)$, encoding the geometric delay directly. At the Bessel nulls, the jinc function crosses zero and the visibility phase undergoes a π shift; however, since the amplitude is near zero at these nulls, the SNR is too low to determine the phase reliably there, and any phase-based baseline method (such as Method 2, §4.2) could be corrupted by these jumps.

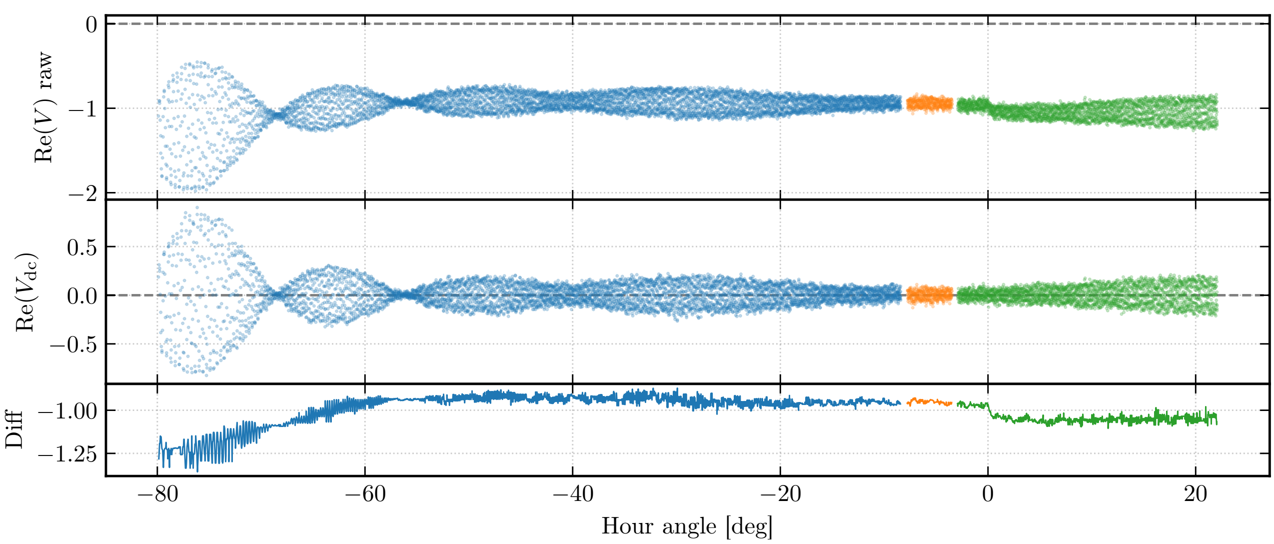

Fig. 5.— DC correction at a representative channel. *Top:* Raw $\mathrm{Re}(V)$. *Middle:* DC-corrected $\mathrm{Re}(V_{\mathrm{dc}})$, now symmetric about zero. *Bottom:* Difference (subtracted DC component). The amplitude drop near $h \approx 0°$ is likely due to a strong wind gust momentarily blowing a dish off-pointing.

4. Baseline Determination

We apply five independent methods to determine the baseline $(b_{\mathrm{ew}}, b_{\mathrm{ns}})$ from the fringe data: (1) STFT fringe-frequency fitting, (2) phase-slope regression, (3) delay-spectrum IFFT, (4) nonlinear least-squares complex fringe fitting, and (5) brute-force 2-D grid search.

4.1. STFT Fringe-Frequency Fit (Method 1)

We slide windowed segments across the time series data, extract the local fringe frequency $\hat{f}_f(h)$ from the FFT peak in each segment, and fit the resulting $\hat{f}_f(h)$ curve to equation (7) via least squares. This is performed per chip (to avoid windowing across gaps) with a prior-constrained peak search. The method recovers both b_{ew} and b_{ns} and provides the most intuitive visualisation of the fringe (Figure 6). This method is included in the adopted baseline.

4.2. Phase Slope (Method 2)

The phase of the complex visibility varies linearly across frequency with slope $\mathrm{d}\phi/\mathrm{d}\nu = 2\pi\,\tau_g(h)$ (Figure 7). The unwrapped phase is linearly regressed to extract delay coefficients, from which b_{ew} and b_{ns} can be obtained. This method is systematically biased for solar observations because the Bessel-function nulls introduce π phase jumps that corrupt the unwrapping.

4.3. Delay Spectrum (Method 3)

The delay spectrum is the inverse FFT (IFFT) of the cross-correlation spectrum $V_{12}(\nu)$, transforming from the frequency domain to the delay domain. The peak of the delay spectrum at each capture directly yields the relative delay $\tau(h)$. The delay curve is then fitted to the physical model of Equation (3). This broadband, frequency-independent estimator is robust against Bessel-null phase jumps (since it operates in the delay domain rather than the phase domain) and is included in the adopted baseline.

Figure 8 shows the measured delay versus hour angle with the best-fit model overlaid. The sinusoidal sweep of $\tau(h)$ directly captures all three key interferometric quantities: the amplitude of the sinusoidal variation gives b_{ew}, the $\cos h$ modulation gives b_{ns}, and the vertical offset (the fit intercept at $h = 0$ $\tau_{\mathrm{inst}} = 47.2$ ns) gives the effective cable delay τ_c'; at transit ($h = 0$), the dominant EW term vanishes and the measured delay reads off τ_c' directly (see §4.7).

4.4. Nonlinear Least-Squares Fringe Fit (Method 4)

We fit the fringe equation (6) directly to both the real and imaginary parts of the complex visibility via the model

$$V_i = (A + iC)\cos\psi_i + (B + iD)\sin\psi_i, \qquad (13)$$

where $\psi_i = 2\pi\nu\,\tau_g'(h_i)$ is the geometric fringe phase at hour angle h_i, with τ_g' given by Equation (3). The nonlinear parameters $(b_{\mathrm{ew}}, b_{\mathrm{ns}})$ are iterated, while the linear parameters (A, B, C, D) are solved algebraically at each iteration. Fitting both real and imaginary components simultaneously provides $2N$ data points and the smallest uncertainties of any method. This method is included in the adopted baseline.

4.5. Brute-Force Grid Search (Method 5)

Since for any trial baseline $(Q_{\mathrm{ew}}, Q_{\mathrm{ns}})$ the geometric delay $\tau_g'(h_i)$ is fixed, the linear coefficients (A, B) can be solved numerically by ordinary least squares. Hence, by evaluating the sum-of-squared residuals

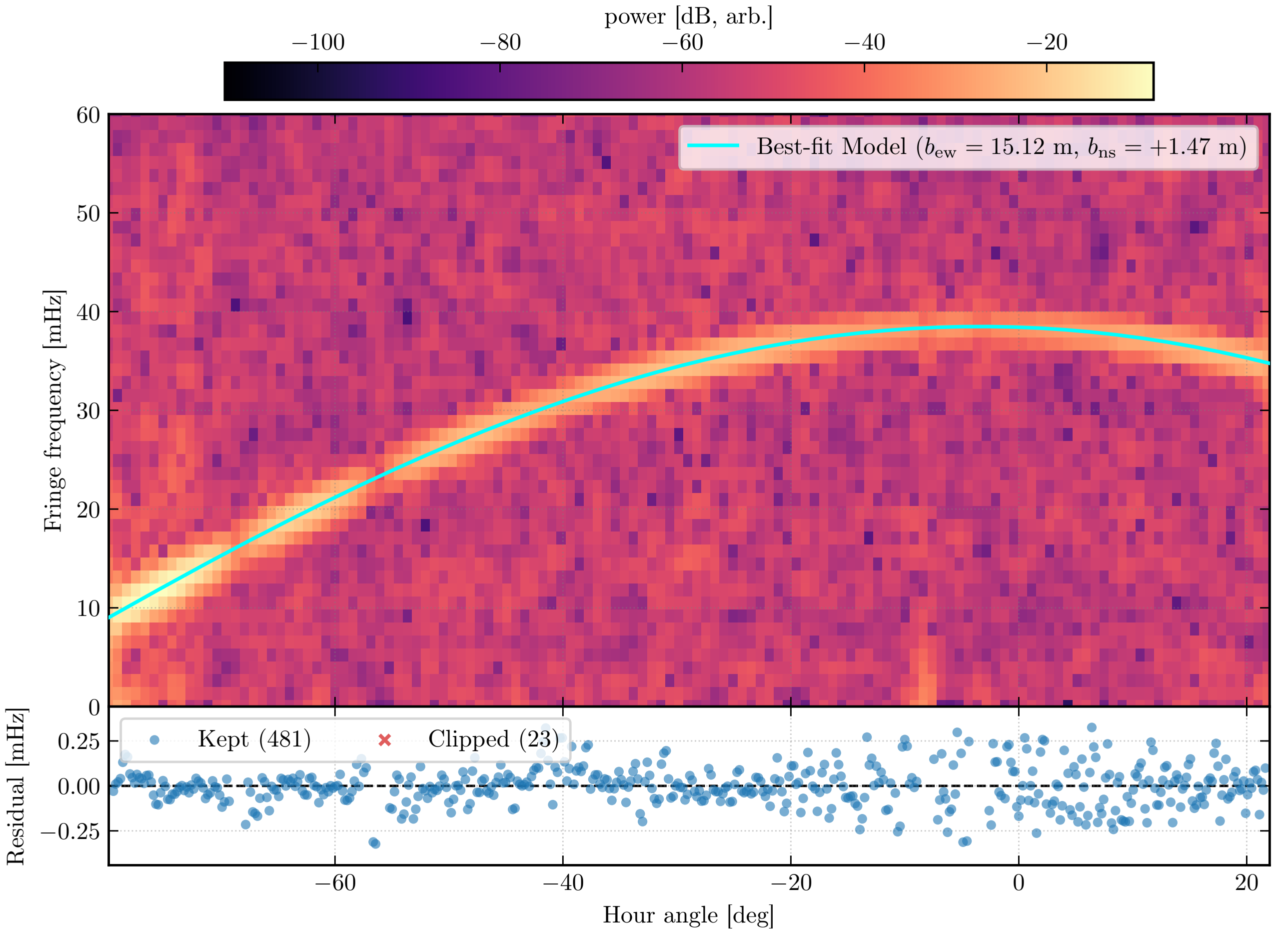

Fig. 6.— STFT fringe-frequency fit (Method 1). *Top:* Spectrogram of the windowed FFT power versus hour angle, with the best-fit model $f_f(h) = \omega_\oplus[(b_{\rm ew}/\lambda)\cos\delta\,\cos h - (b_{\rm ns}/\lambda)\sin L\,\cos\delta\,\sin h]$ overlaid (cyan curve; $b_{\rm ew} = 15.12$ m, $b_{\rm ns} = 1.47$ m). The bright ridge traces the instantaneous fringe frequency across the track. *Bottom:* Fit residuals after sigma-clipping; 481 points are retained and 23 are clipped.

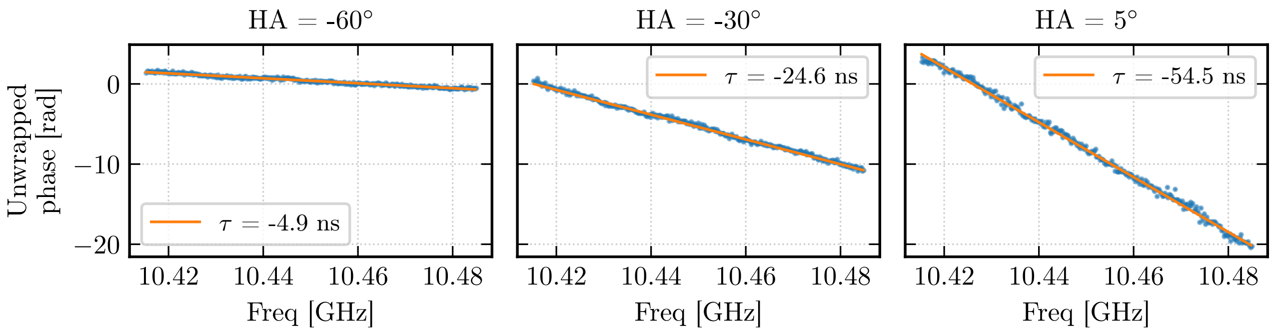

Fig. 7.— Phase slope across frequency at three hour angles. The linear slope recovers τ_g directly; fitted delays agree with predictions to within a few nanoseconds.

$\mathcal{S}^2(Q_{\rm ew}, Q_{\rm ns})$ on a grid around some prior baselines, we can find the best-fit values by minimization.

Following A. Parsons (2017), the residual surface near the minimum is approximated by the second-order Taylor expansion

$$\Delta\mathcal{S}^2 \;\equiv\; \mathcal{S}^2 - \mathcal{S}^2_{\rm min} \;=\; \Delta\mathbf{Q}^{\sf T}\,[\alpha]\,\Delta\mathbf{Q}, \qquad (14)$$

where $\Delta\mathbf{Q} = (Q_{\rm ew} - Q^*_{\rm ew},\ Q_{\rm ns} - Q^*_{\rm ns})^{\sf T}$ and $[\alpha]$ is the 2×2 curvature matrix whose elements are the numerical second derivatives $\frac{1}{2}\partial^2\mathcal{S}^2/\partial Q_i\,\partial Q_j$ evaluated at the minimum. The covariance matrix is $[\epsilon] = [\alpha]^{-1}$, and the 1σ uncertainties are $\sigma_{Q_i} = \sqrt{\epsilon_{ii}}\cdot\sigma$, where $\sigma^2 = \mathcal{S}^2_{\rm min}/{\rm dof}$.

The confidence contours are drawn at the multi-

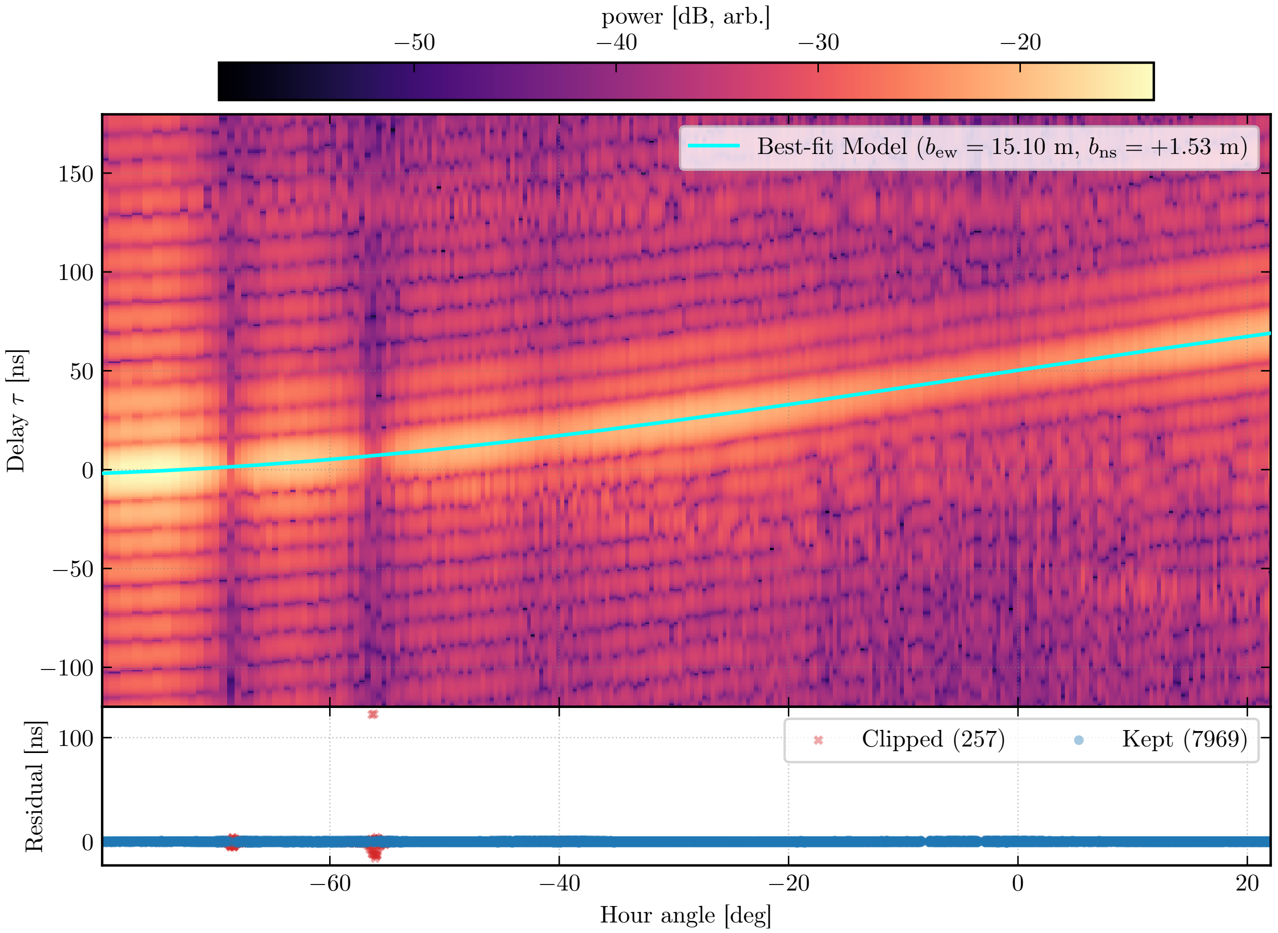

Fig. 8.— Delay spectrum versus hour angle (Method 3). Each point is the delay extracted from the IFFT of the cross-correlation spectrum $V_{12}(\nu)$ at a single capture. The solid curve is the best-fit model $\tau(h) = (b_{ew}/c)\cos\delta\,\sin h + (b_{ns}/c)\sin L\,\cos\delta\,\cos h + \tau_{inst}$. The sinusoidal amplitude varies by b_{ew}; the vertical offset at $h = 0$ is the effective cable delay τ'_c.

plicative level $2.30 \times \chi^2_{\nu,\min}$, corresponding to the $\Delta\chi^2$ threshold for 68.3% confidence with two parameters of interest[2]; these contours are shown for illustration only, and the adopted grid-search uncertainties are instead taken from the analytical curvature matrix [ε]. Figure 9 shows the 2-D χ^2 surface with these contours.

The brute-force model fits a point-source fringe (Equation 6) and does not account for the Bessel-function envelope that modulates the resolved solar visibility. At large $|h|$ the projected baseline b_{proj} changes rapidly, resulting in varying visibility ampltiudes; the envelope therefore imprints a slow amplitude modulation that biases the least-squares residuals to only minimize for large $|h|$. Restricting the fit to a symmetric window $|h| \leq 20°$ around transit mitigates this since near transit the projected baseline varies slowly and the Bessel envelope is nearly flat. Hence, the point-source fringe model is a better approximation to the data. Accordingly, the windowed search yields b_{ew} = 15.114 m, closer to the adopted baseline than the full-track result of b_{ew} = 15.103 m (Figure 9).

[2]For a χ^2 distribution with $p = 2$ dof, $\Delta\chi^2 = -2\ln(1-0.683) = 2.30$.

4.6. Adopted Baseline Value

Table 2 compares all five methods. The adopted baseline is the mean of the three methods included in the adoption set (STFT, NLS-complex, and delay-spectrum), with per-method errors added in quadrature and the inter-method spread (sample standard deviation) added in quadrature as a systematic, yielding $b_{ew} = 15.11 \pm 0.01$ m and $b_{ns} = 1.49 \pm 0.05$ m. The uncertainties are sub-centimetre, reflecting the high SNR of the fringe signal across 285 spectral channels and >8000 time samples.

As a direct validation, Figure 10 shows the recovered fringe pattern near transit from the windowed grid search minimum. The best-fit point-source sinusoid tracks the observed Re(V) oscillations at the ∼26 s fringe period, with residuals broadly in agreement with radiometric noise. This confirms that the baseline solution correctly predicts the interferometric fringe rate and phase, and that the point-source model is an excellent description of the data in the near-transit regime where the Bessel envelope is flat.

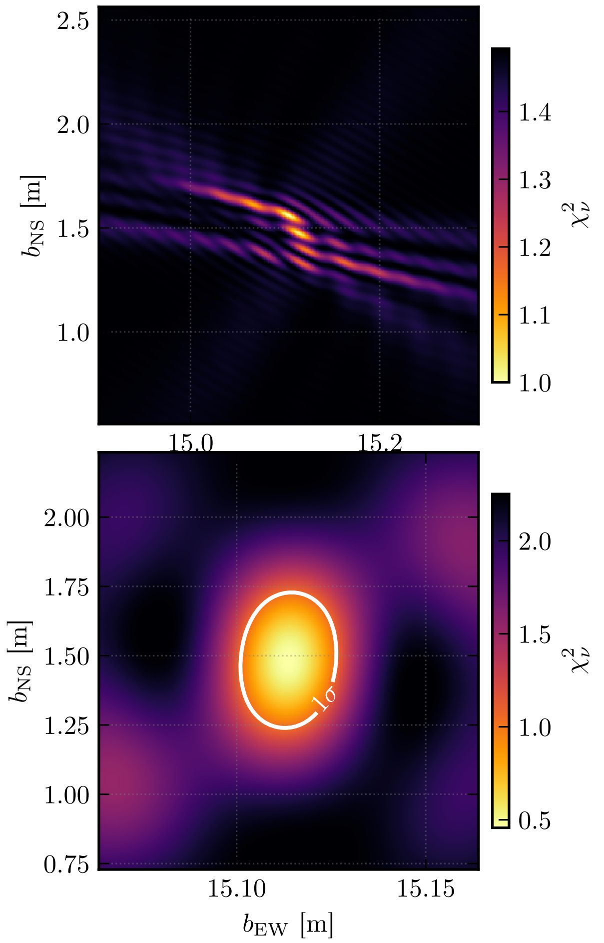

Fig. 9.— 2-D brute-force χ^2_ν grid search in the $(b_{\rm ew}, b_{\rm ns})$ plane. *Top:* Full-track search using all hour angles. *Bottom:* Windowed search restricted to $|h| \leq 20°$ around transit, suppressing Bessel-envelope bias. The 1σ confidence contour ($2.30\times$ the minimum χ^2_ν) is shown for illustration; adopted uncertainties come from the analytical curvature matrix. The windowed result ($b_{\rm ew} = 15.114$ m) is closer to the adopted baseline than the full-track grid ($b_{\rm ew} = 15.103$ m), consistent with a small systematic from the envelope modulation at large $|h|$.

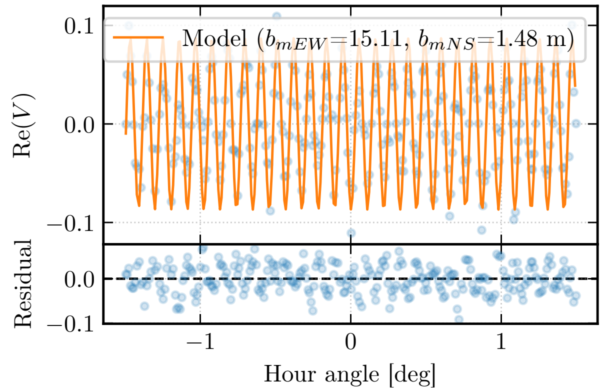

Fig. 10.— Recovered fringe pattern near transit ($|h| \leq 1.5°$) from the windowed grid search. *Top:* observed Re(V) (scatter) and best-fit sinusoidal model (solid), illustrating the rapid fringe oscillation resolved by the interferometer. *Bottom:* fit residuals.

4.7. Effective Cable Delay

At transit ($h = 0$), the dominant east–west geometric delay vanishes ($\sin h = 0$), and the total measured delay reduces to $\tau_{\rm tot}(h = 0) \approx 3.05$ ns $+ \tau'_c$.

Next, we extract τ'_c two ways. First, fitting the phase slope $\mathrm{d}\phi/\mathrm{d}\nu = 2\pi\,\tau_{\rm tot}$ across the 92 captures within $\pm 0.5°$ of transit and subtracting the known NS geometric term gives $|\tau'_c| = 52.2 \pm 0.6$ ns. Second, the constant term from the Method 3 delay fit over the entire hour-angle range gives $\tau_{\rm inst} = 47.2$ ns. The two estimates differ by $\sim$5 ns in magnitude and have opposite signs, reflecting the delay sign convention (which antenna's signal arrives first). The magnitude difference likely arises from residual fringe structure at transit or possible imperfect separation of the NS geometric term. The equivalent cable path-length difference is $|\tau'_c| \times c \approx 15.6$ m.

5. Solar Diameter Measurement

5.1. Geometric Prior from Extrema Positions

Six Bessel extrema are identified per channel directly from the DC-corrected visibility using automatic peak/valley detection within pre-computed

Table 2: Baseline comparison: all five methods. Columns show the east–west and north–south baseline components and their uncertainties. The LiDAR measurement is shown for reference.

Method	$b_{\rm ew}$ [m]	$\sigma_{b_{\rm ew}}$ [m]	$b_{\rm ns}$ [m]	$\sigma_{b_{\rm ns}}$ [m]
STFT fringe-freq	15.1152	0.0028	1.4659	0.0063
Phase slope	14.5948	0.0002	1.7558	0.0008
Delay spectrum	15.1048	0.0306	1.5307	0.0946
NLS (complex)	15.1148	0.0000	1.4746	0.0000
Grid (full)	15.1031	0.0002	1.5634	0.0005
Grid (±20°)	15.1136	0.0001	1.4837	0.0023
Adopted (mean)	**15.11**	**0.01**	**1.49**	**0.05**
LiDAR measurement	15.2	0.3	1.4	0.3

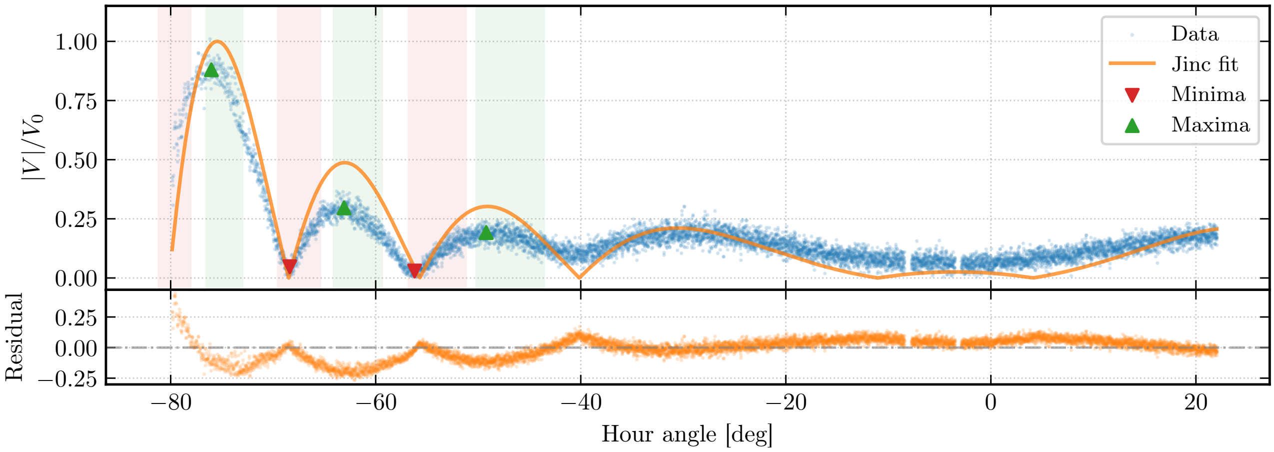

Fig. 11.— Jinc fit and Bessel extrema at a representative channel. The scatter shows the normalised $|V|$ versus hour angle; the orange curve is the best-fit jinc model with R free. Triangles mark identified maxima (orange) and minima (red); colored shaded regions denote the hour-angle ranges where extrema are expected for chromospheric diameters $D \sim 32$–$35'$. The fitted diameter annotation appears in the upper right.

Table 3: Solar diameter comparison. Literature radio diameters are converted from published radii ($D = 2R$); uncertainties are propagated accordingly.

Method / Reference	Frequency	Diameter [arcmin]	Note
Extrema mean (this work)	10.45 GHz	34.66 ± 0.03	statistical only
Jinc fit (this work)	10.45 GHz	34.46 ± 0.01	R bounded by $\pm 3\sigma$ prior
Optical photosphere	—	$\sim$31.97	T. M. Brown & J. Christensen-Dalsgaard (1998)
Radio	9 GHz	32.97 ± 0.07	F. Menezes & A. Valio (2017)
Radio	11 GHz	33.03 ± 0.17	F. Menezes & A. Valio (2017)
Radio	17 GHz	32.55 ± 0.05	C. L. Selhorst et al. (2004)
Radio	18–26 GHz	32.6–32.8	M. Marongiu et al. (2024)

hour-angle windows: three maxima (at the J_2 zeros $j_{2,1}$, $j_{2,2}$, $j_{2,3}$) and three minima (at the J_1 zeros $j_{1,1}$, $j_{1,2}$, $j_{1,3}$).

We obtain a prior for the solar radius from the positions of these extrema rather than fitting the full envelope shape. This is more robust because the presence of sunspots prevents the Bessel nulls from reaching zero amplitude, biasing any least-squares fit to the jinc. By contrast, the locations of the extrema in $|u|$-space depend only on R through the Bessel zeros (Equation 11), independent of the sunspot floor or limb brightening.

Each identified extremum yields an independent solar radius estimate via $R_k = j_k/(2\pi\,|u_k|)$, where $|u_k|$ is the measured projected baseline at the extremum's hour angle. Per-channel radii are combined via an unweighted mean across all matched extrema, with errors added in quadrature, $D_{\rm mean} = 34.66 \pm 0.03$ arcmin.

5.2. Standard Jinc Fit

We verify the extrema-based result by fitting the jinc function $A\,|2J_1(2\pi|u|R)/(2\pi|u|R)|$ to the band-averaged visibility envelope with R as a free parameter, constrained to the $\pm 3\sigma$ range from the extrema prior. Figure 11 overlays the best-fit jinc model together with the identified Bessel extrema on the raw $|V|$ scatter.

Table 3 summarises the measured and literature solar diameters. The extrema-based mean gives $D = 34.66 \pm 0.03'$, while the Jinc fit with R free returns $D = 34.46 \pm 0.01'$; the two estimates differ by $\sim$0.2', well within the systematic uncertainties discussed in §6.1. Both values are $\sim$5% larger than the $\sim$33.0' expected at 10.5 GHz by interpolating the 9 GHz and 11 GHz single-dish measurements compiled by F. Menezes & A. Valio (2017), and $\sim$8% larger than the optical photosphere (T. M. Brown & J. Christensen-Dalsgaard 1998). Because the literature references bracket our observing frequency, the frequency-dependent chromospheric limb height cannot account for the excess; the discrepancy is instead attributed to the systematics described in §6.1.

6. Discussion

At centimetre wavelengths, the solar emission is dominated by thermal free–free (bremsstrahlung) radiation from the chromosphere and transition region, with brightness temperatures $T_b \sim 10^4$ K at 10 GHz (G. A. Dulk 1985). Because the free–free opacity scales as ν^{-2}, the $\tau = 1$ surface lies at progressively greater heights at longer wavelengths, and the radio Sun is expected to be a few percent larger than the optical photosphere.

The uniform-disk model predicts that the visibility amplitude should vanish at the Bessel nulls, but in our results, the observed nulls do not reach zero. Two physical effects are to blame. With the aforementioned larger expected solar radii, with limb darkening, the radial profile deviates from a uniform disk in a way that the visibility amplitude near the nulls is elevated above zero. Secondly, active sunspot regions contribute compact, bright radio emissions. Therefore, an unresolved sunspot will add a constant visibility floor that fills in the nulls. SDO/HMI continuum images from the observation date (2026 March 20) confirm the presence of an active region on the disk.

6.1. Systematic Uncertainties

The excess in our results cannot be explained by the chromospheric height gradient and limb brightening alone; both the baseline determination and the diameter measurement carry systematic uncertainties that dominate over their statistical errors.

For the baseline, the dominant systematic is the Bessel-envelope modulation of the fringe amplitude: at hour angles where the visibility passes through a null, the effective SNR drops and the fringe phase becomes poorly defined, biasing methods that assume a constant-amplitude sinusoid. The phase-slope method is most severely affected, compounded with π phase jumps at the nulls corrupt the unwrapping which together can be blamed for the $\sim$0.5 m outlier seen in Table 2. For the other four non-outlier methods, they are more resistant to this systematic and agree broadly at the centimetre level.

For the diameter, the uncertainties ($\sim$0.03' from extrema, $\sim$0.01' from the Jinc fit) are negligibly small; the measurement is instead limited by how well the baseline is known and by departures from the uniform-disk model. Since the angular diameter scales inversely with the baseline length, any fractional baseline error propagates directly into the diameter. From the DC-offset removal process, any residual calibration errors will also distort the visibility envelope and shift the apparent positions of the Bessel extrema. Together with the aforementioned physical effects, collectively they plausibly account for the $\sim$5% excess of our measured diameter over published values at comparable frequencies.

6.2. Comparison of Baseline Methods

Excluding the phase-slope outlier, the four remaining methods (STFT, delay-spectrum, NLS-complex, and brute-force grid) agree in $b_{\rm ew}$ to within $\sim$0.01 m but show a much larger spread of $\sim$0.1 m in $b_{\rm ns}$ (Table 2). This asymmetry is geometric. From Equation (3), the geometric delay has a $b_{\rm ew}$-dependence in $\cos\delta\,\sin h$, whereas $b_{\rm ns}$ couples with $\sin L\,\cos\delta\,\cos h$, making it even-symmetric about transit. Moreover, the h-independent part of the NS delay is absorbed into the effective cable delay τ_c' (Equation 4), so $b_{\rm ns}$ is partially degenerate with the instrumental delay. Our asymmetric hour-angle coverage (-80° to $+22^\circ$) further limits the $\cos h$ available to break this degeneracy. This is especially evident in the brute-force grid search, where the χ^2 surface is elongated along the $b_{\rm ew}$–$b_{\rm ns}$ diagonal with secondary sidelobes, symptomatic of the covariance between the two components (Figure 9).

Among the four non-outlier methods, the STFT and NLS-complex fits yield the tightest uncertainties and agree to within 0.001 m in $b_{\rm ew}$ and 0.01 m in $b_{\rm ns}$. The delay-spectrum method, which estimates the baseline from the peak of the inverse Fourier transform of the cross-power spectrum, has the largest errors ($\sigma_{b_{\rm ns}} \approx 0.09$ m) because the spectral resolution of the IFFT limits how finely the delay can be localised. The full-track brute-force grid returns a slightly lower $b_{\rm ew}$ and higher $b_{\rm ns}$ than the other methods, consistent with the Bessel-envelope bias discussed above; restricting the grid to $|h| \leq 20^\circ$ brings it into closer agreement.

7. Conclusions

From a single horizon-to-horizon drift scan of the Sun at 10.45 GHz with the NCH two-dish interferometer, we recover the baseline geometry, instrumental cable delay, and solar structure.

Five independent baseline methods yield an adopted baseline of $b_{\rm ew} = 15.11 \pm 0.01$ m and $b_{\rm ns} = 1.49 \pm 0.05$ m, consistent with an independent LiDAR measurement at the 0.3σ level. The east–west component is constrained to sub-centimetre precision across all non-outlier methods, while the north–south component exhibits an order-of-magnitude larger scatter ($\sim$0.1 m), a consequence of the weaker $\cos h$ geometric leverage, its partial degeneracy with the cable delay τ_c', and the asymmetric hour-angle coverage of our track (-80° to $+22^\circ$). The brute-force χ^2 surface confirms this covariance, with confidence contours elongated along the $b_{\rm ew}$–$b_{\rm ns}$ diagonal. Two independent measurements of the effective cable delay—from the phase slope at transit ($|\tau_c'| = 52.2 \pm 0.6$ ns) and the delay-spectrum intercept (47.2 ns)—differ by $\sim$5 ns, reflecting the difficulty of separating the small NS geometric contribution.

The solar diameter, derived from the Bessel-function modulation of the visibility envelope, is $D_\odot = 34.66 \pm 0.03'$ from the extrema-based method and $34.46 \pm 0.01'$ from a jinc fit. Both values exceed published radio diameters at comparable frequencies by $\sim$5%; the excess is dominated by systematic baseline uncertainty rather than a chromospheric height effect. Non-zero visibility at the Bessel nulls is consistent with limb brightening and an unresolved active region, the latter confirmed by SDO/HMI imagery from the observation date.

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A. Reproducibility Notes

All code, notebooks, cached analysis products, and report sources used for this lab are version-controlled at `https://github.com/raw-asparagus/ay-121`.